WHOLESALE MARKET MANAGEMENT

MULTI-AGENT AI ORDERING PLATFORM

PRODUCT REQUIREMENTS DOCUMENT | VERSION 2.0 | 21 OCTOBER 2026

Status: Proposed final product specification • Hackathon-first delivery • Architecture handoff ready

Prepared from partner PRD v1.0, with explicitly proposed extensions for voice, multi-agent workflows, resilience and demo scope.

1. Executive summary

Build a mobile-first B2B wholesale marketplace connecting independent wholesalers with shopkeepers (clients). Each wholesaler owns a separate catalog, inventory, sales, costs and receivables. Shopkeepers can discover and purchase goods from several wholesalers through normal checkout or a WhatsApp-inspired AI chat. The assistant accepts typed messages and recorded speech, especially Pashto, Urdu and English, and responds with both readable text and generated speech in the detected or selected language. A Super Admin oversees onboarding, dispatch, audits and platform reporting; delivery assignees receive only relevant shipment access.

Specialized AI agents assist discovery, conversation, cart drafting, stock checking, notifications and analytics. The server remains the source of truth: AI may propose actions but cannot bypass authorization, invent product data, approve its own orders, or silently post money movements.

2. Scope, assumptions and priority

Priority	Commitment	Delivery definition
P0 / hackathon	Working end-to-end core	Auth/roles; wholesaler product and stock; client AI text + Pashto/Urdu/English voice attempt; cart confirmation; multi-wholesaler split; approvals; admin dispatch; simple payments and demo tests.
P1 / stretch	Polished and reliable	Spoken AI replies for supported languages, persistent chat, rich UI, receipts, notifications, basic analytics, retries and local draft persistence.
P2 / post-hackathon	Production hardening	OTP/SMS, live delivery GPS, payment gateway, advanced returns, credit policy, sophisticated risk models, integrations and comprehensive observability.

P0 is a minimum coherent demonstration, not permission to skip security or data integrity. Voice recognition/synthesis quality must be tested on actual Pashto/Urdu examples before promising universal language coverage. “Any language” is best-effort, not a guaranteed acceptance criterion.

3. Human roles and access control

Role	Can do	Cannot do
Super Admin (one designated account)	Invite/suspend wholesalers; see platform KPIs; inspect scoped records; assign deliveries; view audit events; handle exceptions.	See plaintext passwords; erase historic payments; make unaudited financial corrections.

Wholesaler (many independent tenants)	Manage own catalog, cost, prices, stock, approvals, invoices, receivables and business dashboard.	See or mutate another wholesaler's private records; administer other wholesalers.
Shopkeeper / Client (many)	Register; browse public listings; chat with AI; prepare/confirm orders; view own suborders, delivery and balances.	See other clients' private records or wholesalers' internal costs/profits.
Delivery assignee (restricted operational user)	View assigned shipment, contact/delivery details; update pickup, failure and delivery proof.	View unrelated shipments, costs, platform finances or inventories.

A shopkeeper is a human actor. AI agents are backend services, never substitutes for authenticated human roles. Delivery access can be a restricted mobile page rather than a fourth full dashboard in P0.

4. Core product experience

4.1 Shopkeeper: mobile-first conversational ordering

- Primary screen: WhatsApp-like conversation with message bubbles, send/record controls, recording preview, playback, timestamps, pending/error states, chat history and product cards.
- Language: typed or spoken Pashto, Urdu, English and mixed phrases; auto-detect with manual override; preserve original transcript and allow editing before order submission.
- Response: same-language text and, when synthesis succeeds, playable same-language voice. If voice fails, show text and an honest retry/fallback indicator; never block checkout.
- Search and comparison: only live authorized catalog results, displaying wholesaler, SKU/brand, pack size, unit, price, availability and minimum quantity.
- Cart: conversational add/change/remove, clarification for ambiguous quantities, packaging and supplier; one consolidated preview with per-wholesaler subtotals, delivery address and explicit confirmation.
- Secondary UI: conventional catalog, cart, order timeline, invoices and balance views, accessible without using AI.

4.2 Wholesaler operations

- Unique wholesaler tenant, private login, dashboard and product management: SKU, aliases (Pashto/Urdu/English), unit, pack size, cost, selling price, minimum order quantity, reorder point, stock and status.
- Review only relevant lines/suborders; approve/reject with reason; reserve approved quantities atomically; see pending, reserved, fulfilled and available quantities.
- Review own revenue, captured cost of goods sold, gross margin, receipts and customer debt; download simple report (P1).

4.3 Admin and delivery operations

- Admin creates/invites/suspends wholesalers; sees global order and dispatch queues, stock alerts, exceptions and scoped audited financial views.
- Only approved suborders can be dispatched; admin assigns a delivery assignee and records expected date/reference.
- Delivery assignee updates assigned status and proof; failed deliveries and returns trigger exception review, not silent stock or payment changes.

5. AI agent architecture: functional contract

The orchestration layer routes requests to bounded specialist agents/tools. These are logical responsibilities and may run inside one backend process for the hackathon; they do not require six separate LLM instances.

Agent / service	Responsibility	Allowed outputs and boundaries
Conversation Orchestrator	Detect intent/language, maintain context, route tasks, ask clarifying questions.	Replies, tool requests; no direct database mutation.
Speech & Language Agent	Speech-to-text, transcript editing flow, multilingual response drafting, text-to-speech.	Transcript, language, text, audio URL; handle unavailable synthesis gracefully.
Catalog Discovery Agent	Resolve aliases, SKU, brand, packaging; compare available offers.	Only products returned by authorized catalog queries; never fabricate prices/stock.
Order Draft Agent	Build and modify cart, split by wholesaler, calculate preview and ask confirmation.	Draft updates only; order creation via authenticated confirmed server endpoint.
Inventory Validation Agent	Check stock and packaging, detect shortages and conflicts.	Read-only suggestions; reservation occurs transactionally in server order workflow.
Business Insights Agent	Summarize authorized wholesaler/admin sales, receivables and low stock.	Read-only scoped aggregates; no ledger edits.
Workflow/Notification Agent	Generate contextual event messages and recommended next actions.	Send notifications from verified backend events; no self-approved payments/dispatch.

Trust model: model output is untrusted. Every tool receives the authenticated actor, tenant scope, schema validation, rate limit and audit correlation ID. All monetary arithmetic, stock changes and order state transitions are deterministic server logic.

6. Canonical business lifecycle

1. Shopkeeper browses or chats, reviews exact products and builds a draft cart.
2. Client confirms a final quote (products, wholesaler, unit/pack, quantity, locked unit price, subtotal, delivery). Server creates a parent order and per-wholesaler suborders/items using an idempotency key.
3. Each wholesaler reviews only its own suborder lines; approval atomically reserves available stock; rejected items carry reasons and do not affect other wholesalers.
4. Approved suborders enter admin dispatch queue; admin assigns a delivery assignee and tracking reference.
5. Assignee records pickup, out-for-delivery, delivered/failed and proof; stock is consumed at the defined fulfillment transition; canceled reservations are released.
6. System creates or finalizes distinct invoices per wholesaler for fulfilled items; authorized user records cash/bank payments with method, reference and collector; allocations reduce that wholesaler's receivable.
7. Return, failure or correction creates explicit reversal/credit and audit events; never rewrite prior payment history.

Parent order states: Draft → Submitted → In Review → Partially Approved / Approved / Rejected → Partially Fulfilled / Fulfilled → Closed, plus Cancelled. Each suborder and shipment has its own state. Payment states are independent: Unpaid / Partially Paid / Paid / Refunded. A rejected suborder cannot be dispatched.

7. Business rules and edge cases

- One checkout can include multiple wholesalers, but each line belongs to exactly one wholesaler; separate suborders, invoices, ledgers and dispatch permissions.
- At checkout, revalidate active product, current price, pack size, MOQ and delivery address; if changed, require renewed human confirmation.
- Quantity uses decimal-safe units; support kg, bag, carton, piece, liter and explicit pack conversions. "5 bags" must not be interpreted as "5 kg"; unknown bag weight requires clarification.

- Concurrency: lock/reserve stock in a transaction; available = on_hand − reserved; never negative. Expired or canceled reservations must be released exactly once.
- Retries: idempotency keys for order submission, payments, dispatch updates and stock adjustments; network errors must not create duplicates.
- Voice errors: unsupported codec, failed transcription, mixed languages, silence and ambiguous names preserve audio/transcript for retry or manual editing; no order submission from raw unreviewed transcript.
- AI conversation must not expose hidden cost, private contact records or other tenant information, including through prompt injection.
- If wholesaler is suspended, block new approvals and catalog sales while retaining historic records. If delivery fails, record a reason and require an authorized retry/return action.
- Payment: record partial/full cash or transfer; no payment is considered cleared until verified by an authorized collector/admin. Refunds and credits are separate auditable entries.
- Financials: gross profit = recognized net sales − captured cost of goods sold; report cash collected, sales, receivables and profit separately; no profit on unapproved orders.

8. Minimum data model

Domain	Core entities / fields
Identity	users, roles, wholesaler_memberships, clients, delivery_assignees, sessions; status, tenant ownership.
Catalog	wholesalers, products, product_aliases, units/packaging, price history, inventory_movements, reservations.
Orders	carts, cart_items, orders, suborders, order_items, status_events, approvals; price/cost snapshots.
Logistics	shipments, assignments, delivery_events, proof attachments and exception reasons.
Finance	invoices, invoice_lines, payments, payment_allocations, credit_notes, reversals, ledger_events.
AI	conversations, messages, audio_assets, transcripts, AI_tool_calls, AI_action_audit, language preferences.
Operations	notifications, audit_events, idempotency_keys, retry records and error events.

Database must use foreign keys, unique constraints, monetary decimal types, transaction boundaries, scoped indexes and immutable audit/ledger entries. For P0, simplify tables only when tenant isolation and accounting traceability remain intact.

9. Security, privacy and resilience

- Backend-enforced RBAC and tenant isolation on every endpoint, export, audio upload and AI tool. Client-provided wholesaler_id is never trusted as authorization.
- Password hashing, secure sessions/tokens, restricted uploads, request validation, login throttling, secure secrets and HTTPS in deployed environments.
- Sensitive voice files and chat transcripts have access control and a defined retention/deletion policy; never place API keys or private financial data in browser prompts.
- Audit who created, approved, rejected, assigned, delivered, posted, reversed or accessed sensitive cross-tenant data.
- Persist draft messages/cart locally where feasible; show connectivity and retry states; replay pending writes only with idempotency keys.
- Backups, structured errors, correlation IDs and a basic health check. For production, add automated recovery, monitoring, encrypted storage and disaster recovery procedures.

10. UX, accessibility and performance

- Mobile-first responsive experience; chat is primary for shopkeepers, operational tables and dashboards are optimized for desktop and mobile.
- Readable Pashto/Urdu right-to-left rendering and English left-to-right rendering, mixed-script support, keyboard navigation and visible labels.
- Show loading/recording/transcribing/generating/sending states; avoid blocking on TTS; provide clear fallback when AI unavailable.
- All displayed money in PKR; store timestamps in UTC and show Pakistan Standard Time in UI; paginate large lists.
- Performance targets (proposed, validate in environment): standard catalog query p95 under 2 seconds on test data; non-AI API actions p95 under 1 second; voice/LLM responses are asynchronous with progress and timeout feedback.

11. Hackathon implementation sequence

Stage	Build and demo checkpoint
M1 — Foundation	Database, seed three wholesalers and shopkeepers, auth/RBAC, tenant isolation, sample catalog and stock.
M2 — Commerce	Client product search/cart, per-wholesaler split orders, approval/rejection, reservation and status timeline.
M3 — Dispatch + finance	Admin queue, assigned delivery status, one invoice per wholesaler, partial payment and balances.
M4 — AI text	Chat UI, conversation persistence, catalog tools, draft modifications, clear confirmation gate.
M5 — Voice	Record → transcribe → editable preview → AI reply in same language → TTS where supported.
M6 — Verification + demo	Run acceptance tests, fix critical issues, record a complete multilingual multi-wholesaler journey.

Deadline guardrail: stop adding features once the end-to-end demo works. If voice output in Pashto is unreliable, retain text and playback controls with a visible limitation; do not fabricate a successful speech demo.

12. Acceptance criteria (must test)

ID	Test / expected result
AT-01	Admin creates and suspends wholesaler; old records remain; suspended account cannot operate.
AT-02	Wholesaler A cannot read/update wholesaler B records by URL, API or AI tool.
AT-03	Client A cannot view Client B order, chat, voice or payment history.
AT-04	Pashto, Urdu and English sample typed requests return same-language text; speech input is transcribed and editable.
AT-05	If same-language voice synthesis succeeds, response audio is playable; if not, visible text fallback works.
AT-06	Ambiguous bag/brand/size prompts clarification; AI displays only live matching offers.
AT-07	AI add/edit/remove cart works across turns; only explicit final confirmation creates an order.
AT-08	Mixed-wholesaler cart creates correct scoped suborders and invoices; each owner sees only own lines.
AT-09	Concurrent approvals cannot oversell; rejected/cancelled lines release reservations correctly.
AT-10	Admin assigns approved shipment; delivery user accesses

	only assigned shipment; timeline logs status.
AT-11	50% verified payment leaves 50% due on the correct wholesaler invoice; retry does not duplicate it.
AT-12	Disconnect/retry preserves draft and does not duplicate confirmed orders.
AT-13	Prompt injection asking for other wholesaler costs or unauthorized payment is denied.
AT-14	A complete shopkeeper → wholesaler → admin → delivery → payment demo can be executed with seed data.

13. Decisions to finalize before implementation

Decision	Recommended hackathon default
Delivery fee / tax / discounts	Zero or fixed disclosed fee; tax and discounts disabled unless explicitly configured.
Payment confirmation	Admin or authorized wholesaler manually verifies payment; no gateway.
Invoice trigger	Create invoice for fulfilled/delivered quantities, not merely submitted draft.
Cancellation / returns	Before dispatch: authorized cancellation and reservation release; after delivery: admin exception and credit note.
Delivery role	Restricted authenticated assignee view; no full independent portal needed.
Client identity	Email/password or phone/password for demo; OTP as production enhancement.
Voice provider	Select after validating real Pashto/Urdu transcription and speech synthesis on target devices.
Scope of autonomous agents	Assist, retrieve and propose; deterministic server actions require human confirmation/authorization.

14. Out of scope / phase 2

Direct WhatsApp integration, real-time phone calling, live GPS, fully automated payment gateway, production SMS OTP, sophisticated fraud scoring, advanced returns/refunds, external accounting integrations, multilingual coverage guarantees beyond tested languages, predictive procurement and multi-region deployment.

15. Architecture handoff checklist

- Confirm tech stack, deployment target and database; map RBAC and tenant boundaries.
- Draw client, wholesaler, admin and delivery interfaces and API contracts.
- Define order/suborder/line-item state machines and inventory/payment transaction boundaries.
- Choose AI model(s), speech provider, tool schemas, prompt-injection defenses, timeout/retry policies and cost limits.
- Prepare seeded demo products with Pashto/Urdu aliases and realistic unit/pack conversions.
- Write end-to-end test fixtures for cross-tenant denial, partial payment, voice review and idempotent submission.

Source alignment: Sections 1, 3, 4.2, 4.3, 6, 7, 8 and 9 retain and clarify the original v1.0 business model. Expanded conversational UI, mandatory multilingual experience, multi-agent contracts, resilience, prioritization and detailed edge-case acceptance tests are proposed v2.0 additions; defaults in Section 13 require stakeholder approval.